

Shivam Shetty

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🐙 GitHub | 🔗 LinkedIn | 🏆 Kaggle

SUMMARY

AI/ML undergraduate with experience in AI/ML and full-stack systems. IEEE published co-author, NAIN 2.0 grant recipient, and multiple-time hackathon winner skilled in developing and evaluating ML models and deploying AI and full-stack applications using Python, FastAPI, Next.js, Docker and edge hardware.

PUBLICATIONS

Enhancing STEAM Education Using Virtual Technologies: An Augmented Reality and Virtual Reality-Based Approach

[View Paper](#)

IEEE Xplore, 2026.

Swasthik Yesh, Manvitha Ravi Salian, **Shivam Shetty**, Gowri Bhat.

SKILLS

Languages & DBs: Python, SQL, TypeScript, JavaScript, C, Java, PostgreSQL, MongoDB

AI, ML & Data: PyTorch, LangChain, MONAI, TensorFlow, scikit-learn, Pandas, NumPy, ONNX, TinyML

Full-Stack & DevOps: FastAPI, Next.js, Django, React, Docker, Git, Firebase, Supabase

CS Concepts: DSA, OOP, DBMS, RESTful APIs

Tools: Streamlit, Postman, Roboflow, Weights & Biases

ACHIEVEMENTS AND ACTIVITIES

Leadership & Core Activities

- **Technical Lead – Aikya Club (Dept. of AIML, SMVITM)** **2026**
 - Organize hackathons, coding contests, and AI/ML workshops; mentor student development and project teams.
- **Full-Stack Collaborator – Varnothsava 2026 Website** **2026**
 - Co-developed a Next.js, TypeScript, and Tailwind CSS platform handling **20+ events** and **500+ participants**.

Awards & Recognition

- **Product Funding Selection** – Secured over Rs. 4 Lakhs in funding under the **NAIN 2.0 program** for the project PluckSmart to drive commercial product development.
- **1st Place** – MedhaDrishti National Level AI Hackathon (Medical Image Enhancement & Segmentation), YUGMA TechFest 2.0, JNNCE Shivamogga [\[View Certificate\]](#)
- **1st Place** – MONAITHON Hackathon, JNNCE, Shivamogga [\[View Certificate\]](#)
- **1st Place** – Algorithm Roulette ML Competition, College Tech Fest [\[View Certificate\]](#)
- **Honorary Mention** – IEEE Eu-reka 2026, Pune Section [\[View Certificate\]](#)
- **3rd Place** – HackYugma Hackathon, YUGMA TechFest, JNNCE [\[View Certificate\]](#)
- **Best Project** – Lost and Found Portal (First-Year Mini Project) [\[View Certificate\]](#)

PROJECTS

Khandava – Satellite-Based Forest Fire Prediction

[\[GitHub\]](#)

- Engineered GDAL-based preprocessing pipelines to integrate multi-source geospatial and meteorological data from **NASA (FIRMS)**, **ISRO**, and **ECMWF (ERA5)**.

- Optimized an **XGBoost** model to handle extreme class imbalance (450k+ samples), achieving **75% Recall** for prioritized early fire detection.

Edge AI Beehive Monitoring (Honorary Mention – IEEE Eu-reka 2026)

[\[GitHub\]](#)

- Built an **ESP32-based IoT system** integrating DHT22, HX711, and microphone sensors for real-time hive monitoring.
- Developed **MFCC-based audio classification** for bee-sound analysis and optimized the pipeline for lightweight edge inference.
- Integrated hive and weather datasets and developed **feature engineering and anomaly detection** pipelines using Isolation Forest to identify abnormal environmental conditions.

Medical Image Enhancement and Segmentation (1st Place – MedhaDrishti)

- Developed an **Auto-Encoder, Residual U-Net, U-Net** model using **MONAI** and **PyTorch** on BraTS and custom Brain MRI image data securing **1st Place** out of all competitors.

Medical Image Segmentation (1st Place – MONAITHON)

[\[GitHub\]](#)

- Developed a **U-Net** model using **MONAI** and **PyTorch** for automated retinal vessel extraction on the DRIVE dataset, securing **1st Place** out of all competitors.
- Deployed a production-ready, low-latency inference application using **FastAPI**, **Streamlit**, and **Docker**.

Animal Sound Recognition – Audio Classification

[\[GitHub\]](#)

- Developed an audio classification pipeline using **YAMNet**, **MFCCs**, and **Mel spectrograms** for animal sound recognition.
- Achieved **0.846 Recall** and optimized the pipeline for **TinyML/edge deployment**.

PluckSmart Coffee Ripeness Detection & Custom Dataset Engineering

[\[GitHub\]](#)

- Curated and annotated a custom dataset of **3,000+ high-resolution images** (Roboflow) to capture real-world environmental noise and lighting variations.
- Trained and fine-tuned a **YOLO** object detection model for high-precision classification of ripe vs. unripe coffee cherries.

Department Voting Platform (120+ Users)

[\[GitHub\]](#)

- Built and deployed a secure **Next.js** and **Firebase** web application handling **120+ voters** with real-time vote casting.
- Implemented role-based access control and **Firestore transactions** to prevent duplicate voting and ensure data integrity.

EDUCATION

2023 – 2027	B.E. in Artificial Intelligence and Machine Learning Shri Madhwa Vadiraja Institute of Technology (<i>CGPA: 8.2</i>)	Udupi, KA
2021 – 2023	Pre-University Course (PUC) Don Bosco Senior Secondary School (<i>Percentage: 84.8%</i>)	Shirva-Udupi, KA
2021	Secondary School Leaving Certificate (SSLC) Don Bosco English Medium School (<i>Percentage: 79.6%</i>)	Shirva-Udupi, KA